Pricing the Battlefields

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Submission: **1, 4, 6, 1, 7, 28, 3, 5, 33, 12**

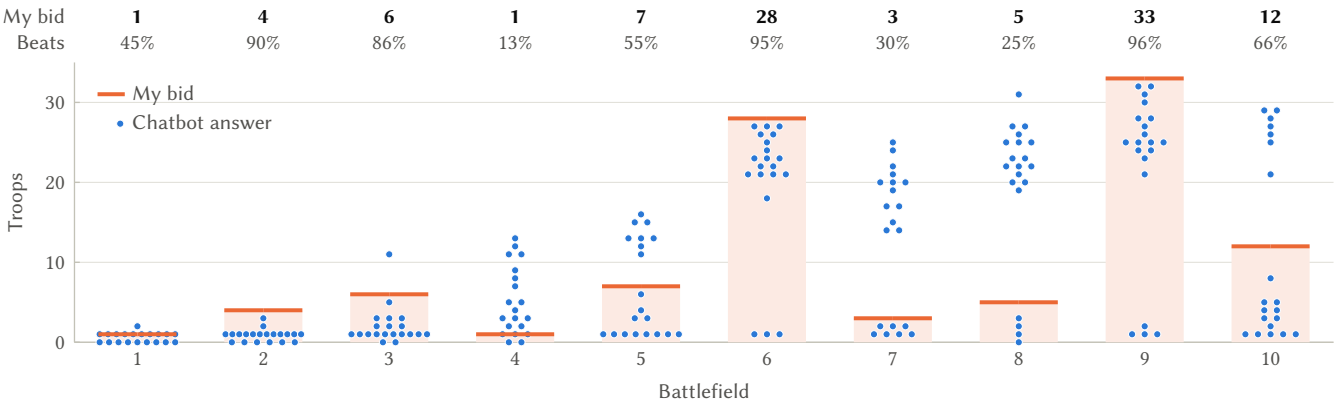


Figure 1: My bids (orange line; the light bar shows the troops spent) against 20 chatbot answers (blue dots). A dot above my line beats me on that battlefield, a dot on it ties, and a dot below it loses. The header rows give my troops and the share of the whole simulated field that each bid beats.

Summary. I target battlefields 2, 3, 5, 6, 9 and 10, three adjacent pairs whose values sum to 35, so winning all six fires every multiplier and scores 80.85. I bid one troop more than the highest of 20 chatbot answers on 6 and 9 and give up 7 and 8, where most chatbots spend 14 to 31 troops each. Against a simulated field weighted toward chatbot answers, the allocation beats 85% of opponents and finishes in the top quartile by wins and by points in all 1,000 mock tournaments.

1 What the scoring rewards

Each player wins one of $2^{10} = 1{,}024$ sets of battlefields, so I precompute every score and store it as an integer times 100, which makes ties exact. Four facts shape the game.

1. The multipliers compound. If every won battlefield has a won neighbor, the base is a multiple of 5, and the set contains 6, the score is $1.5 \times 1.4 \times 1.1 = 2.31$ times the base.
2. Rule 2 depends on values modulo 5. Battlefields 5 and 10 never change it, and the pairs $\{1, 4\}$, $\{2, 3\}$, $\{6, 9\}$ and $\{7, 8\}$ cancel.
3. Winning more can score less. In 556 of the 5,120 ways to add one battlefield to a set, the score falls: $\{6, \ldots, 10\}$ scores 92.4, but $\{1, 6, \ldots, 10\}$ scores 67.1.
4. You can win a match while losing six battlefields. Twenty-six sets of four outscore the other six; $\{2, 6, 8, 9\}$ beats its complement 51.59 to 50.4. No set of three or fewer does.

The pairs $\{2, 3\}$, $\{5, 6\}$ and $\{9, 10\}$ use the first two facts at once. Their values sum to 35, and winning all six scores $35 \times 2.31 = 80.85$, 64% of the maximum 127.05 from 35 of the board's 55 points. Battlefields 5 and 10 are the cheap ones to lose, because the base stays a multiple of 5: $\{2, 3, 6, 9, 10\}$ still scores 64.68, and $\{2, 3, 5, 6, 9\}$ scores 50.82.

2 The field

No Blotto allocation wins in isolation, so I model the other applicants. AI use is allowed, and I expect most applicants to ask a chatbot. I collected 20 answers by hand, 10 from Claude and 10 from ChatGPT, each from a fresh chat given the problem verbatim. Most agree on prices (Figure 1): 18 to 27 troops on battlefield 6, 21 to 32 on 9, 14 to 31 on 7 and 8 when they contest them, and at most 3 on 1 and 2.

Table 1: The simulated field. Range is the share of the field each group may take, and center is its midpoint, rescaled so the shares sum to 100%. Win rate is my pick's, against each group with its jittered copies.

Group	Entries	Range	Center	Win rate
Chatbot answers	20	70–95%	73%	88.5%
Riddler entries	4,510	0–20%	9%	70.2%
Rules of thumb	44	0–15%	7%	66.6%
Random splits	1,000	0–5%	2%	72.9%
Counters	30	5–15%	9%	86.7%

I model the field as five groups (Table 1). The group weights are judgment calls, so each group gets a range; the center mix uses the midpoints, and the selection step assumes the worst weights inside the ranges. Human play comes from 4,510 entries to FiveThirtyEight's Battle for Riddler Nation, which used the same board without the multipliers. Rules of thumb cover even splits, splits proportional to value, all-in on the top k battlefields, and contiguous blocks whose base is a multiple of 5. Random splits draw troops from a Dirichlet-multinomial. Counters stand in for applicants who think a step ahead: six rounds of fictitious play, each adding the five best responses to the pool so far. Every group except the random one carries a jittered copy of each entry, with 1 to 3 troops moved once or twice and 30% of the group's weight, so the search cannot exploit one exact number.

3 The allocation

Each battlefield has a price list: the share of the field that a given number of troops beats. I buy where the list jumps and stop one troop past the jump.

- **6 and 9.** 28 and 33 troops top every chatbot answer and beat 95% and 96% of the whole field.
- **2 and 3.** Ten troops in total beat 90% and 86%, because 38 of the 40 chatbot bids on them are 3 troops or fewer.
- **5 and 10.** 7 and 12 troops beat 55% and 66%. Losing either costs its face value, never the 1.4 multiplier.

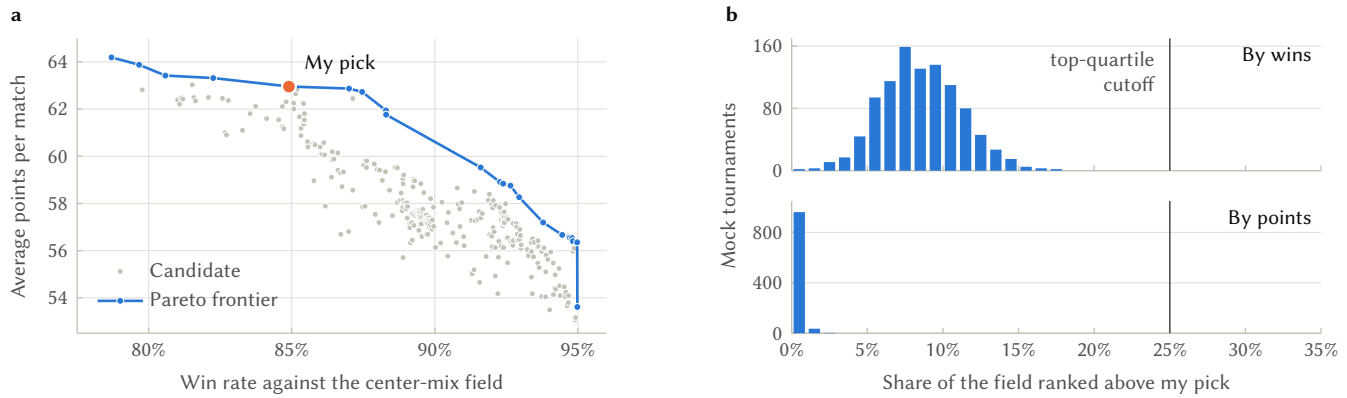


Figure 2: (a) Win rate and average points per match for the 300 candidates. My pick sits at the knee of the frontier. (b) My pick's rank in 1,000 mock tournaments, as the share of the field ranked above it. All 1,000 ranks fall left of the top-quartile cutoff.

Table 2: My most common outcomes against the center-mix field.

Battlefields I win	Base	Score	Matches
2, 3, 5, 6, 9, 10	35	80.85	17.7%
1, 2, 3, 5, 6, 9, 10	36	59.40	7.5%
2, 3, 6, 7, 9, 10	37	61.05	7.4%
2, 3, 6, 9	20	34.65	4.4%
2, 3, 6, 9, 10	30	64.68	4.3%
2, 3, 5, 6, 9	25	50.82	4.1%

- **7 and 8.** 3 and 5 troops concede the two battlefields where the chatbots overpay, yet still beat the 30% and 25% of opponents who abandon them.
- **1 and 4.** One troop each ties the many one-troop bids.

Table 2 lists my most common outcomes. The full target set leads at 18% of matches. The next two each win one extra battlefield, 1 or 7, which breaks the multiple of 5 and costs about 20 points. The troop on battlefield 1 still earns its place, because ties there decide matches. Against the ChatGPT answer 1, 1, 3, 4, 1, 1, 21, 19, 28, 21, my troop ties theirs and holds their base at 29, so I win 50.82 to 36.5. Without it they also take battlefield 1, their base reaches 30, their 1.4 multiplier fires, and I lose 50.82 to 52.5. Moving that troop to any other battlefield cuts my win rate from 84.9% to 81.9% or less. My won set has a base that is a multiple of 5 in 54% of matches and contains 6 in 95%, against 31% and 51% for the average chatbot answer.

4 Search and robust selection

I rate an allocation x against a weighted field by

$$R_\lambda(x) = \lambda\, W(x) + (1-\lambda)\, P(x)/127.05,$$

where W is the win rate and P the average points per match. The organizers rank on an unknown mix of wins and points, so λ is unknown as well.

I search with simulated annealing over integer allocations, using 32 chains of 3,000 steps that move 1 to 5 troops at a time, then polish with steepest ascent over every shift of 1 to 3 troops until none helps. I run the search under the center mix and under every extreme mix of the ranges, for $\lambda \in \{0.25, 0.5, 0.75\}$, add two searches that maximize the worst case below, and keep the 300 best distinct candidates.

A pessimist then chooses the group weights w inside the ranges B and $\lambda \in [0.25, 0.75]$ to minimize each candidate's rating, and I submit the candidate with the best worst case:

$$x^\star = \arg\max_x \min_{w \in B,\, \lambda} \sum_g w_g R_{\lambda,g}(x).$$

The rating is linear in w, so a greedy fill gives the pessimist's exact minimum, and the worst λ is always an endpoint. My pick's worst case is 0.574. The four runners-up sit within 0.002 of it and differ from it by at most two troops on any battlefield, so the shape is robust and the last troop or two is noise. Figure 2a shows the trade-off the pessimist forces: the candidates with the most wins beat 95% of the field but score at most 56.4 points per match, while my pick gives up 10 points of win rate to score 63.0.

I considered two alternatives and rejected both. An equilibrium strategy mixes over many allocations, but I submit one allocation into a field that does not play the equilibrium. Reinforcement learning would chase the same objective with more noise: the game is a single move, and I can score any allocation against the whole field in milliseconds.

5 Validation

Mock tournaments. I drew 1,000 fields of 100 to 300 applicants, with group weights uniform inside the ranges, and played full round robins. My pick finished in the top quartile by wins and by points in all 1,000 (Figure 2b). The median share of the field ranked above it is 8.4% by wins, and in 79% of tournaments no applicant scored more total points.

Held-out data. A rerun that never saw ChatGPT's answers or Riddler round 4 chose 2, 4, 4, 6, 2, 30, 6, 6, 31, 9, with the same shape: about 30 troops each on 6 and 9 and single digits elsewhere. It beats 90% of the unseen ChatGPT answers, the same rate as on the Claude answers it trained on, though its points per match fall from 70.8 to 57.5. On Riddler round 4 its win rate slips from 68% to 64%.

Counters. My exact allocation has a counter: 1, 5, 9, 4, 8, 32, 7, 12, 9, 13 beats it, and my pick wins none of 51 matches against it and 50 jittered copies of it. No entry in the simulated field lies within 10 troops of that counter, so beating me this way takes an applicant who targets my numbers.

6 Assumptions and risks

I read the rules this way: battlefields sit on a line, so 1 and 10 are not adjacent; equal totals are a win for neither player; and total points means the sum of my own scores. The chatbot share is my biggest judgment call. If the field looks more like the Riddler humans, the pick still beats 70% of them. The mock tournaments draw from the pool that produced the pick, so they flatter it, and the held-out run gives the fairer estimate.